Classical theories of information science

信息论选讲之一

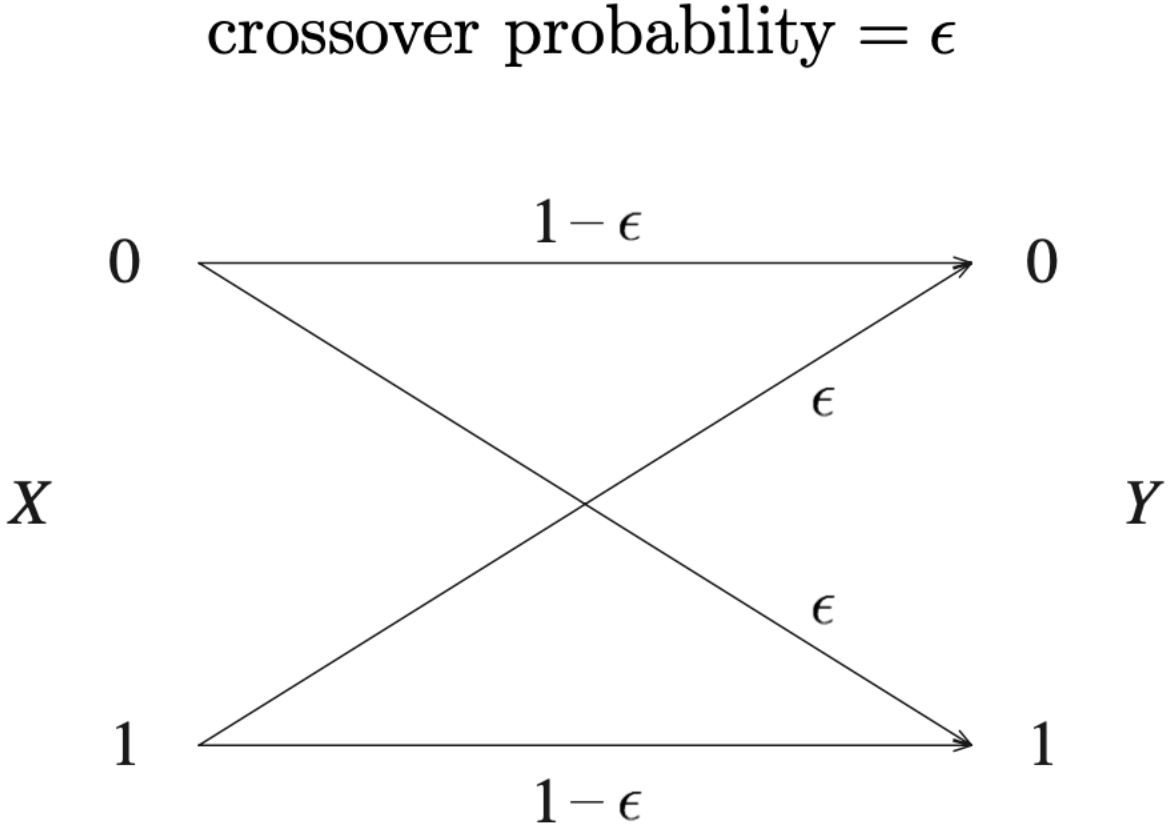
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Outline

1. Definition and Compacity
2. The Channel Coding Theorem
3. The Converse Proof
4. The Direct Proof

1. Definition and Compacity

Binary Symmetric Channel (BSC)



The transition diagram of a binary symmetric channel.

Consider a simple channel code

- Assume that the two messages A and B are equally likely.
- The codeword is then transmitted through the channel.
- Our task is to decode the message based on the output of the channel, and an error is said to occur if the message is decoded incorrectly.

Message set = { A, B }

Message	Codeword
A	0
B	1

$$\Pr\{A|Y=0\} = \Pr\{X=0|Y=0\} \tag{7.1}$$

$$= \frac{\Pr\{X=0\}\Pr\{Y=0|X=0\}}{\Pr\{Y=0\}} \tag{7.2}$$

$$= \frac{0.5(1-\epsilon)}{\Pr\{Y=0\}}. \tag{7.3}$$

Since

$$\Pr\{Y=0\} = \Pr\{Y=1\} = 0.5 \tag{7.4}$$

by symmetry[2], it follows that

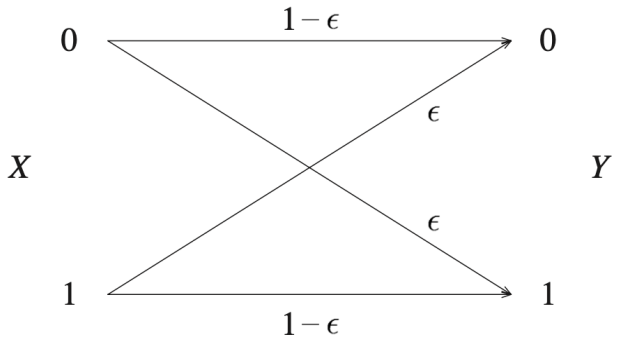
$$\Pr\{A|Y=0\} = 1-\epsilon \tag{7.5}$$

and

$$\Pr\{B|Y=0\} = 1 - \Pr\{A|Y=0\} = \epsilon. \tag{7.6}$$

Since $\epsilon < 0.5$,

$$\Pr\{B|Y=0\} < \Pr\{A|Y=0\}. \tag{7.7}$$



Consider a simple channel code

Therefore, in order to minimize the probability of error, we decode a received 0 to the message A. By symmetry, we decode a received 1 to the message B.

An error occurs if a 0 is received and the message is B, or if a 1 is received and the message is A. Therefore, the probability of error, denoted by P_e, is given by

$$P_e = \Pr\{Y=0\}\Pr\{B|Y=0\} + \Pr\{Y=1\}\Pr\{A|Y=1\} \quad (7.8)$$
$$= 0.5\epsilon + 0.5\epsilon \quad (7.9)$$
$$= \epsilon, \quad (7.10)$$

where (7.9) follows from (7.6) because

$$\Pr\{A|Y=1\} = \Pr\{B|Y=0\} = \epsilon \quad (7.11)$$

by symmetry.

Let us assume that $\epsilon \neq 0$. Then the above scheme obviously does not provide perfectly reliable communication. If we are allowed to use the channel only once, then this is already the best we can do. However, if we are allowed to use the same channel repeatedly, then we can improve the reliability by generalizing the above scheme.

Message set = { A, B }

Message	Codeword
A	0
B	1

Consider a repetition code

For this more general scheme, we continue to denote the probability of error by P_e. Let N_0 and N_1 be the number of 0's and 1's in the received sequence, respectively. Clearly,

$$N_0 + N_1 = n. \tag{7.12}$$

For large n, if the message is A, the number of 0's received is approximately equal to

$$E[N_0|A] = n(1-\epsilon) \tag{7.13}$$

and the number of 1's received is approximately equal to

$$E[N_1|A] = n\epsilon \tag{7.14}$$

with high probability by the weak law of large numbers. This implies that the probability of an error, namely the event $\{N_0 < N_1\}$, is small because

$$n(1-\epsilon) > n\epsilon \tag{7.15}$$

with the assumption that $\epsilon < 0.5$. Specifically,

Message set = { A, B }

Message	Codeword
A	00…0
B	11…1

n

Consider a repetition code

$$\begin{aligned}
\Pr\{\text{error}|A\} &= \Pr\{N_0 < N_1|A\} && (7.16)\\
&= \Pr\{n - N_1 < N_1|A\} && (7.17)\\
&= \Pr\{N_1 > 0.5n|A\} && (7.18)\\
&\le \Pr\{N_1 > (\epsilon+\phi)n|A\}, && (7.19)
\end{aligned}$$

where

$$0 < \phi < 0.5 - \epsilon, \qquad (7.20)$$

so that ϕ is positive and

$$\epsilon + \phi < 0.5. \qquad (7.21)$$

Note that such a ϕ exists because $\epsilon < 0.5$. Then by the weak law of large numbers, the upper bound in (7.19) tends to 0 as $n \to \infty$. By symmetry, $\Pr\{\text{error}|B\}$ also tends to 0 as $n \to \infty$. Therefore,

$$P_e = \Pr\{A\}\Pr\{\text{error}|A\} + \Pr\{B\}\Pr\{\text{error}|B\} \qquad (7.22)$$

tends to 0 as $n \to \infty$. In other words, by using a long enough repetition code, we can make P_e arbitrarily small. In this sense, we say that reliable communication is achieved asymptotically.

Message set = { A, B }

Message	Codeword
A	00…0
B	11…1

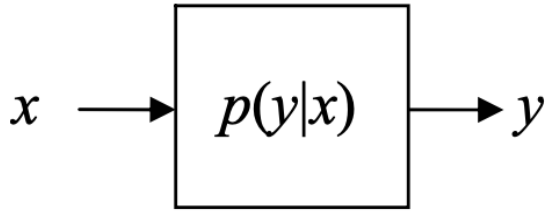
n

Definition of Discrete Channels

Definition 7.1 (Discrete Channel I) Let $\mathcal{X}$ and $\mathcal{Y}$ be discrete alphabets, and $p(y|x)$ be a transition matrix from $\mathcal{X}$ to $\mathcal{Y}$. A discrete channel $p(y|x)$ is a single-input single-output system with input random variable X taking values in $\mathcal{X}$ and output random variable Y taking values in $\mathcal{Y}$ such that

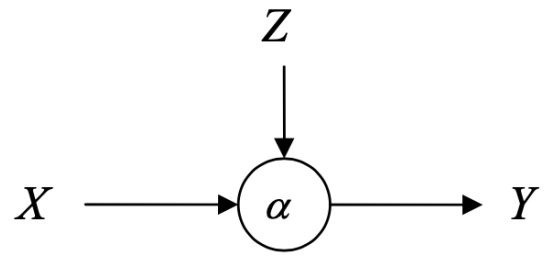
$$\Pr\{X = x, Y = y\} = \Pr\{X = x\}p(y|x)$$

for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$.



Definition 7.2 (Discrete Channel II) Let $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$ be discrete alphabets. Let $\alpha : \mathcal{X} \times \mathcal{Z} \to \mathcal{Y}$, and Z be a random variable taking values in $\mathcal{Z}$, called the noise variable. A discrete channel (α, Z) is a single-input single-output system with input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$. For any input random variable X, the noise variable Z is independent of X, and the output random variable Y is given by

$$Y = \alpha(X, Z).$$



The two definitions are equivalent

- II $\Rightarrow$ I: obvious
- I$\Rightarrow$ II:
  - Define r.v. Z_x with $\mathcal{Z}_x = \mathcal{Y}$ for $x \in \mathcal{X}$ such that $\Pr\{Z_x = y\} = p(y|x)$.
  - Assume Z_x, $x \in \mathcal{X}$ are mutually independent and also independent of X.
  - Define the noise variable $Z = (Z_x : x \in \mathcal{X})$.
  - Let $Y = Z_x$ if $X = x$, so that $Y = \alpha(X, Z)$.
  - Then

$$\begin{aligned}
\Pr\{X = x, Y = y\} &= \Pr\{X = x\}\Pr\{Y = y|X = x\} \\
&= \Pr\{X = x\}\Pr\{Z_x = y|X = x\} \\
&= \Pr\{X = x\}\Pr\{Z_x = y\} \\
&= \Pr\{X = x\}p(y|x)
\end{aligned}$$

Zx is independent of X

Definition 7.3 Two discrete channels $p(y|x)$ and (α, Z) defined on the same input alphabet $\mathcal{X}$ and output alphabet $\mathcal{Y}$ are equivalent if

$$\Pr\{\alpha(x, Z) = y\} = p(y|x)$$

for all x and y.

Some Basic Concepts

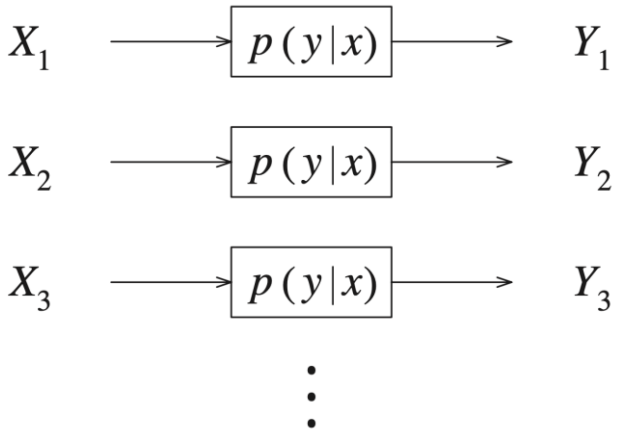
- A discrete channel can be used repeatedly at every time index $i = 1, 2, \ldots$
- Assume the noise for the transmission over the channel at different time indices are independent of each other.
- To properly formulate a Discrete Memoryless Channel (DMC), we regard it as a subsystem of a discrete-time stochastic system which will be referred to as “the system”.
- In such a system, random variables are generated sequentially in discrete-time.
- More than one random variable may be generated instantaneously but sequentially at a particular time index.

Discrete Memoryless Channel (DMC)

Definition 7.4 (DMC I) A discrete memoryless channel (DMC) $p(y|x)$ is a sequence of replicates of a generic discrete channel $p(y|x)$. These discrete channels are indexed by a discrete-time index i, where $i \geq 1$, with the ith channel being available for transmission at time i. Transmission through a channel is assumed to be instantaneous. Let X_i and Y_i be respectively the input and the output of the DMC at time i, and let T_{i-} denote all the random variables that are generated in the system before X_i. The equality

$$\Pr\{Y_i = y, X_i = x, T_{i-} = t\} = \Pr\{X_i = x, T_{i-} = t\} p(y|x)$$

holds for all $(x, y, t) \in \mathcal{X} \times \mathcal{Y} \times \mathcal{T}_{i-}$.



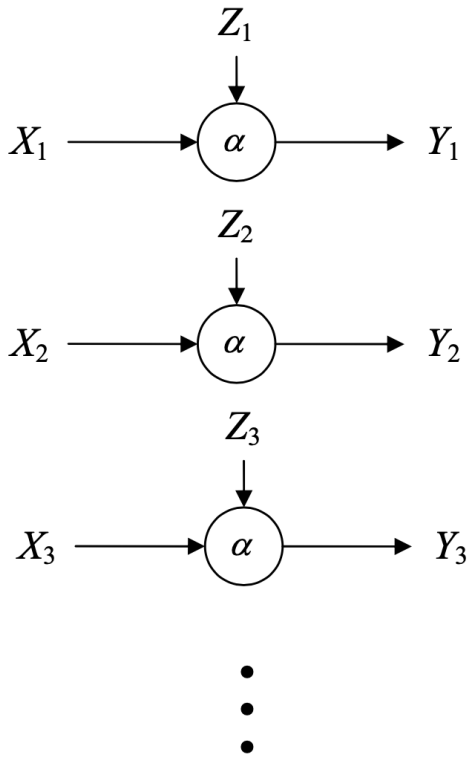
Remark: $T_{i-} \to X_i \to Y_i$, or

Given X_i, Y_i is independent of everything in the past.

Discrete Memoryless Channel (DMC)

Definition 7.5 (DMC II) A discrete memoryless channel (α, Z) is a sequence of replicates of a generic discrete channel (α, Z). These discrete channels are indexed by a discrete-time index i, where $i \geq 1$, with the ith channel being available for transmission at time i. Transmission through a channel is assumed to be instantaneous. Let X_i and Y_i be respectively the input and the output of the DMC at time i, and let T_{i-} denote all the random variables that are generated in the system before X_i. The noise variable Z_i for the transmission at time i is a copy of the generic noise variable Z, and is independent of (X_i, T_{i-}). The output of the DMC at time i is given by

$$Y_i = \alpha(X_i, Z_i).$$



Remark: The equivalence of Definitions 7.4 and 7.5 can be shown. See textbook.

Channel Capacity

Assume both $\mathcal{X}$ and $\mathcal{Y}$ are finite.

Definition 7.6 The capacity of a discrete memoryless channel $p(y|x)$ is defined as

$$C = \max_{p(x)} I(X;Y),$$

where X and Y are respectively the input and the output of the generic discrete channel, and the maximum is taken over all input distributions $p(x)$.

From the above definition, we see that

$$C \geq 0$$

because

$$I(X;Y) \geq 0$$

for all input distributions $p(x)$. By Theorem 2.43, we have

$$C = \max_{p(x)} I(X;Y) \leq \max_{p(x)} H(X) = \log |\mathcal{X}|.$$

Likewise, we have

$$C \leq \log |\mathcal{Y}|.$$

Therefore,

$$C \leq \min(\log |\mathcal{X}|, \log |\mathcal{Y}|).$$

Channel Capacity

Assume both $\mathcal{X}$ and $\mathcal{Y}$ are finite.

Definition 7.6 The capacity of a discrete memoryless channel $p(y|x)$ is defined as

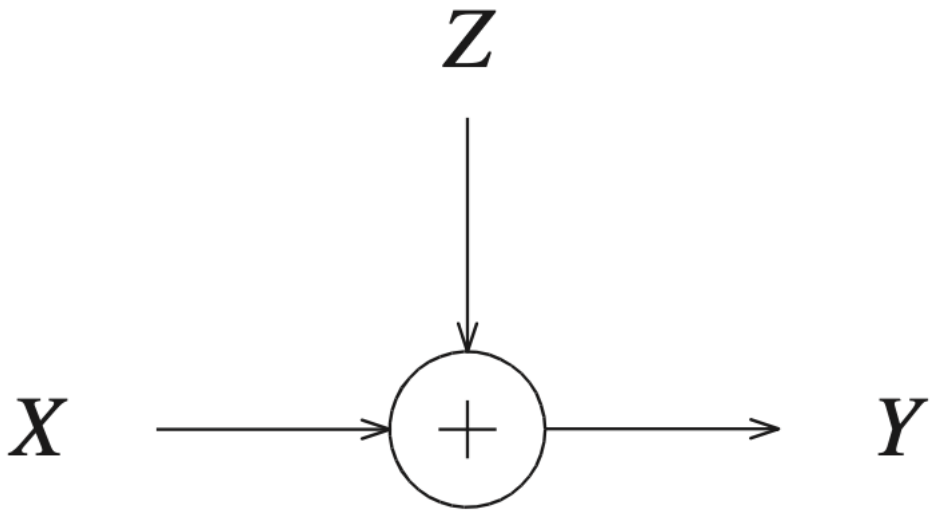
$$C = \max_{p(x)} I(X;Y),$$

where X and Y are respectively the input and the output of the generic discrete channel, and the maximum is taken over all input distributions $p(x)$.

Remarks:

- Since $I(X;Y)$ is a continuous functional of $p(x)$ and the set of all $p(x)$ is a compact set (i.e., closed and bounded) in $\Re^{|\mathcal{X}|}$, the maximum value of $I(X;Y)$ can be attained.
- Will see that C is in fact the maximum rate at which information can be communicated reliably through a DMC.
- Can communicate through a channel at a positive rate while $P_e \to 0$!

Example (BSC)



Alternative representation of a BSC:

$$Y = X + Z \bmod 2$$

with

$$\Pr\{Z = 0\} = 1 - \epsilon \quad \text{and} \quad \Pr\{Z = 1\} = \epsilon$$

and Z is independent of X.

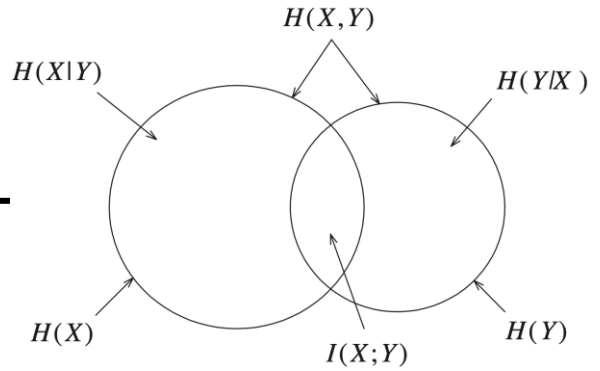
$P(y=1 \mid x=1) = P(Z=0) = 1-\varepsilon$

$P(y=0 \mid x=1) = P(Z=1) = \varepsilon$

$P(y=1 \mid x=0) = P(Z=1) = \varepsilon$

$P(y=0 \mid x=0) = P(Z=0) = 1-\varepsilon$

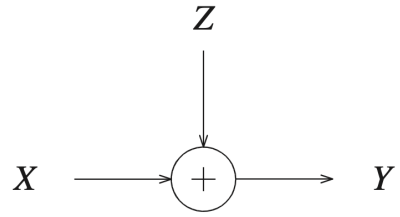
The capacity C of BSC



- In order to determine the capacity of the BSC, we first bound I(X; Y) as follows:

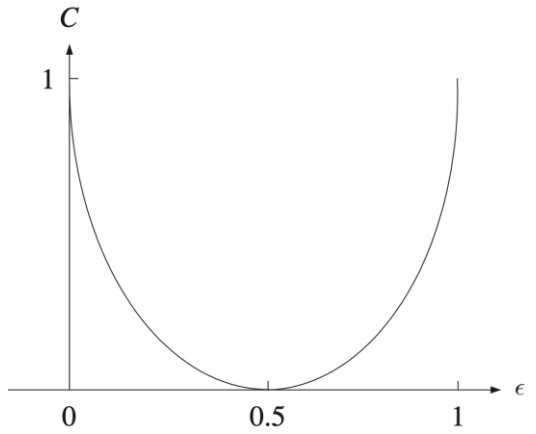
$$\begin{aligned} I(X;Y) &= H(Y) - H(Y|X) \\ &= H(Y) - \sum_x p(x) H(Y|X=x) \\ &= H(Y) - \sum_x p(x) h_b(\epsilon) \\ &= H(Y) - h_b(\epsilon) \\ &\leq 1 - h_b(\epsilon) \end{aligned}$$

where we have used h_b to denote the binary entropy function in the base 2.



$$C = \max_{p(x)} I(X;Y),$$

- So, $C \leq 1 - h_b(\epsilon)$.
- Tightness achieved by taking the uniform input distribution.
- Therefore, $C = 1 - h_b(\epsilon)$ bit per use.



$$h_b(\gamma) = -\gamma \log \gamma - (1-\gamma) \log(1-\gamma)$$

The capacity C of BSC

$$h_b(\gamma) = -\gamma \log \gamma - (1-\gamma)\log(1-\gamma)$$

- C attains the maximum value 1 when $\epsilon = 0$ or $\epsilon = 1$, and attains the minimum value 0 when $\epsilon = 0.5$.
- When $\epsilon = 0$, it is easy to see that $C = 1$ is the maximum rate at which information can be communicated through the channel reliably.
- This can be achieved simply by transmitting unencoded bits through the channel, and no decoding is necessary because all the bits are received unchanged.
- When $\epsilon = 1$, the same can be achieved with the additional decoding step which complements all the received bits.
- By doing so, the bits transmitted through the channel can be recovered without error.

$$C = 1 - h_b(\epsilon)$$

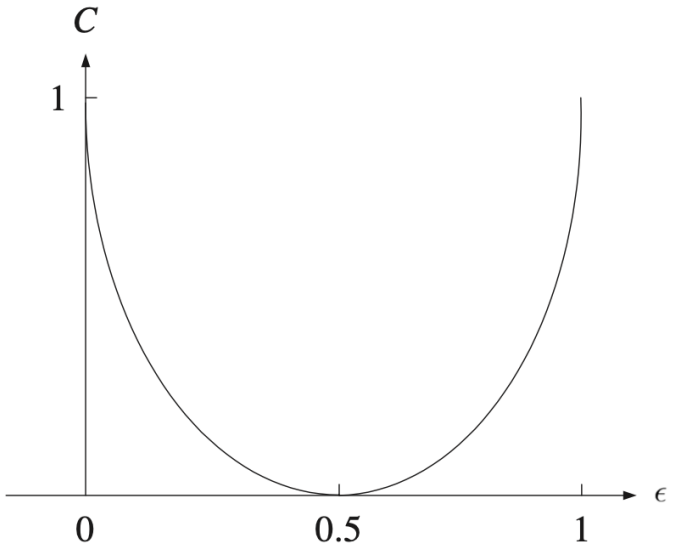
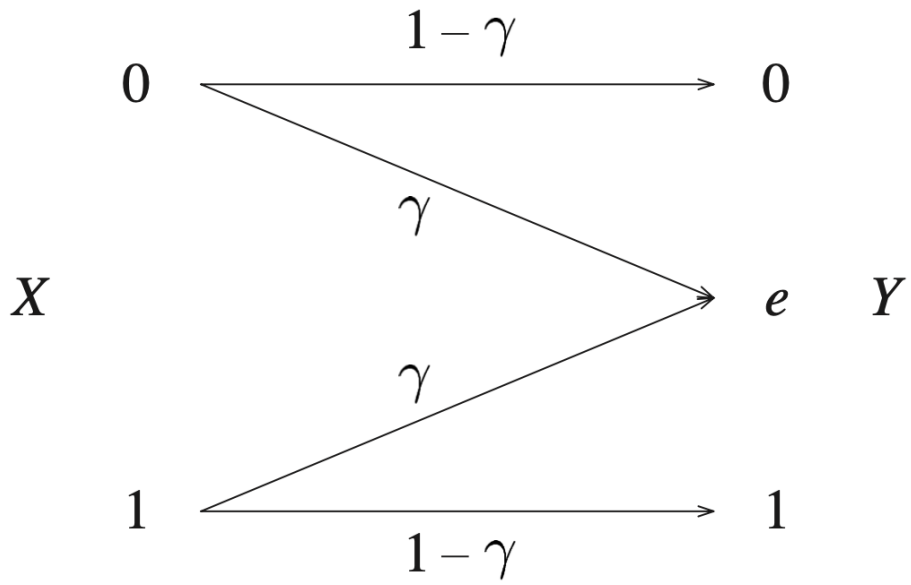


Fig. 7.6. The capacity of a binary symmetric channel.

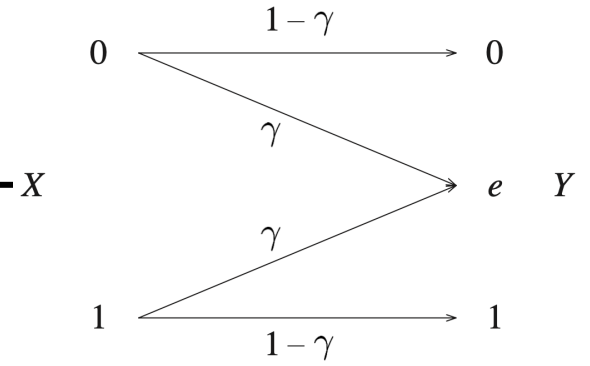
Example (BEC)

- In this channel, the input alphabet is $\{0,1\}$, while the output alphabet is $\{0,1,e\}$.
- With probability γ, the erasure symbol e is produced at the output, which means that the input bit is lost.
- Otherwise the input bit is reproduced at the output without error.
- The parameter γ is called the erasure probability.



Erasure probability $= \gamma$; $C = (1 - \gamma)$ bit per use

The capacity of BEC



To determine the capacity of this channel, we first consider

$$C = \max_{p(x)} I(X;Y) \tag{7.66}$$

$$= \max_{p(x)} (H(Y) - H(Y|X)) \tag{7.67}$$

$$= \max_{p(x)} H(Y) - h_b(\gamma). \tag{7.68}$$

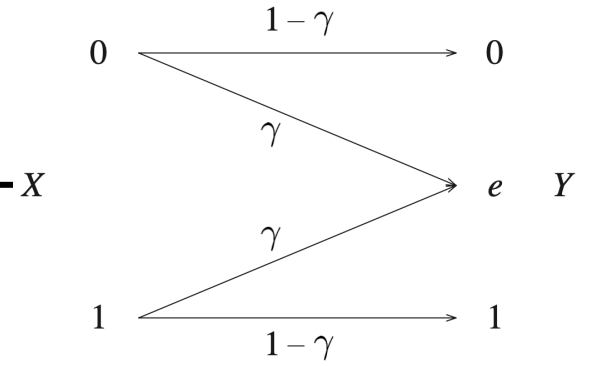
Thus we only have to maximize $H(Y)$. To this end, let

$$\Pr\{X = 0\} = a \tag{7.69}$$

and define a binary random variable E by

$$E = \begin{cases} 0 \text{ if } & Y \neq e \\ 1 \text{ if } & Y = e. \end{cases} \tag{7.70}$$

The capacity of BEC



The random variable E indicates whether an erasure has occurred, and it is a function of Y. Then

$$E = \begin{cases} 0 \text{ if } Y \neq e \\ 1 \text{ if } Y = e. \end{cases}$$

$$\begin{aligned} H(Y) &= H(Y,E) && (7.71)\\ &= H(E) + H(Y|E) && (7.72)\\ &= h_b(\gamma) + (1-\gamma)h_b(a). && (7.73) \end{aligned}$$

$$\Pr\{X = 0\} = a$$

Hence,

$$\begin{aligned} C &= \max_{p(x)} H(Y) - h_b(\gamma) && (7.74)\\ &= \max_{a} [h_b(\gamma) + (1-\gamma)h_b(a)] - h_b(\gamma) && (7.75)\\ &= (1-\gamma) \max_{a} h_b(a) && (7.76)\\ &= (1-\gamma) \text{ bit per use}, && (7.77) \end{aligned}$$

where the capacity is achieved by letting $a = 0.5$, i.e., the input distribution is uniform.

It is in general not possible to obtain the capacity of a DMC in closed form, and we have to resort to numerical computation.

Thank you!